

WATONGA, OK, USA

WMO: 720342

Lat: 35.864N

Lon: 98.421W

Elev: 472

StdP: 95.78

Time Zone: -6.00 (NAC)

Period: 04-19

WBAN: 53947

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-11.0	-7.9	-15.5	1.0	-9.3	-12.7	1.3	-6.1	13.2	11.2	12.1	7.4	4.2	30	0.535

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 11.9	(c) 37.3	(d) 24.2	(e) 36.1	(f) 24.1	(g) 34.7	(h) 24.0	(i) 25.9	(j) 32.9	(k) 25.5	(l) 32.5	(m) 25.0	(n) 31.9	(o) 5.1	(p) 180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
24.3	20.4	28.3	23.8	19.8	28.0	23.1	18.9	27.4	83.3	33.0	80.9	32.9	78.9	31.8	

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 11.8	(b) 10.6	(c) 9.2	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A	(n) N/A	(o) N/A	(p) N/A		
(a) 11.8	(b) 10.6	(c) 9.2	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A	(n) N/A	(o) N/A	(p) N/A		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	15.7	3.7	5.0	11.2	14.5	20.2	26.1	27.4	27.0	23.4	16.2	9.6	3.9		
	DBStd	9.98	5.69	6.35	5.86	5.35	4.53	2.99	3.02	2.91	3.75	5.43	5.50	5.31		
	HDD10.0	727	207	162	57	19	1	0	0	0	0	11	71	199		
	HDD18.3	2054	454	375	229	136	34	0	0	0	6	106	265	449		
	CDD10.0	2821	11	21	93	156	316	483	539	528	402	204	59	9		
	CDD18.3	1106	0	1	7	22	91	233	282	270	158	41	3	0		
	CDH23.3	12248	1	11	62	234	867	2678	3422	3067	1523	359	26	1		
	CDH26.7	5879	0	2	9	62	321	1320	1842	1581	640	100	3	0		
(14) Wind	WSAvg	4.2	4.4	4.6	5.0	5.2	4.2	4.5	3.6	3.3	3.4	4.0	4.4	4.2		
(15) Precipitation	PrecAvg	785	26	32	64	77	101	103	62	99	64	74	47	36		
	PrecMax	1380	68	104	184	201	245	348	144	206	143	191	148	102		
	PrecMin	510	1	0	0	7	19	9	11	1	0	1	0	2		
	PrecStd	196	20	29	50	46	53	80	34	60	40	50	47	31		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	22.4	24.9	27.7	32.0	35.0	37.6	40.2	38.6	35.6	32.0	26.6	20.3		
		MCWB	12.9	15.1	16.4	19.4	21.9	23.8	24.5	24.5	23.2	21.3	18.1	13.6		
	2%	DB	18.4	21.7	25.0	28.0	32.5	36.1	37.3	37.3	33.8	29.3	23.2	17.7		
		MCWB	10.9	13.4	16.6	19.0	22.2	24.0	24.5	24.3	23.5	21.4	16.1	12.9		
	5%	DB	15.9	17.9	22.7	26.1	30.5	34.3	36.0	36.0	32.4	27.3	21.0	14.9		
		MCWB	9.3	12.1	16.1	17.7	21.9	23.6	24.4	24.2	23.1	19.8	15.3	10.1		
	10%	DB	12.9	15.2	20.8	23.8	27.9	32.9	34.5	34.0	30.9	25.1	18.4	12.7		
		MCWB	7.8	10.3	15.4	17.1	21.2	23.4	24.4	24.0	22.6	18.7	14.2	8.4		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	15.1	17.0	20.5	22.3	25.2	26.8	26.8	26.5	25.7	24.0	20.2	17.5		
		MCDB	18.7	20.3	23.8	26.3	30.4	33.1	33.8	33.8	32.3	28.0	23.6	18.2		
	2%	WB	12.0	15.0	19.3	20.9	24.3	25.7	26.0	25.7	24.8	22.9	17.9	13.4		
		MCDB	17.3	19.3	22.6	25.7	29.1	32.4	33.0	32.9	30.8	26.8	20.2	16.0		
	5%	WB	10.0	13.0	17.9	19.7	23.3	25.1	25.5	25.1	24.0	21.8	16.2	10.6		
		MCDB	14.6	18.5	20.5	24.2	28.0	31.8	32.8	32.2	29.5	25.3	19.3	14.3		
	10%	WB	8.2	10.5	16.2	18.5	22.2	24.5	25.0	24.6	23.3	20.2	14.7	8.6		
		MCDB	12.5	14.7	19.5	22.4	26.4	30.9	32.3	31.5	28.4	24.4	18.1	12.4		
(35) Mean Daily Temperature Range	5% DB	MDBR	12.1	12.2	13.0	13.1	11.6	11.6	11.9	11.6	11.4	12.2	12.3	11.2		
		MCDBR	16.3	16.6	14.7	15.4	14.0	13.0	13.5	13.3	12.6	13.5	15.0	15.0		
	5% WB	MCWBR	10.2	9.9	8.7	8.4	6.4	4.5	4.0	4.1	4.5	6.8	9.3	9.8		
		MCWBR	15.0	15.4	11.7	13.2	11.7	11.6	11.9	11.9	10.9	11.0	13.0	13.7		
(40) Clear-Sky Solar Irradiance	taub	taub	0.282	0.291	0.333	0.370	0.398	0.415	0.414	0.417	0.375	0.343	0.303	0.281		
		taud	2.522	2.509	2.407	2.333	2.327	2.321	2.342	2.286	2.379	2.454	2.505	2.526		
	Ebn at Noon	Ebn at Noon	929	958	937	913	887	868	866	857	880	882	893	907		
		Edh at Noon	79	90	110	125	128	128	125	130	112	95	80	74		
(44) All-Sky Solar Radiation	RadAvg	RadAvg	2.84	3.47	4.59	5.68	6.22	6.96	6.94	6.26	5.23	3.94	2.99	2.40		
		RadStd	0.27	0.49	0.32	0.42	0.63	0.56	0.39	0.37	0.37	0.51	0.39	0.25		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (33 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page